OTS PROJECT EXECUTION PROCEDURE

[CONFIDENTIAL]
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Contents

1 Purpose 4
2 Scope 4
3 Abbreviations 4
4 Terms and Definitions 5
5 Responsibilities 5
6 RACI 11
7 SIPOC 15
8 Overall Process Flow Chart 18
9 Process Description 21
10 Information Security requirements 49
11 References 49
12 Annexures 49
13 Forms 56

CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Date Prepared By	Date Reviewed By	Date Approved By
1	Issue B, Version V 0.1	Initial Draft	Implementation of the Integrated Management System	Aniket Parab 12-May-25	Priyanka Shouche 26-May-25	
2	Issue B V 0.2	Updated for Issue B V 0.1 Comments	Updated for Internal Review Comments	Aniket Parab 10-Jun-25	Priyanka Shouche 20-Jun-25	
3	Issue C V 1.0	Updated for Issue B V 0.2 comments	Updated for Internal review comments	Aniket Parab 31-Jul-25	Amogh Vijayadwhaja, Priyanka Shouche 31-JuL-25	Priyanka Shouche, Sanmangala M S 31-Jul-25

1 Purpose

This document defines the standard process for executing an Operator Training Simulator (OTS) project, ensuring consistency, quality, and adherence to industry standards. It outlines the key activities, deliverables, and quality control measures involved in the project lifecycle.

2 Scope

This SOP applies to all OTS projects executed by the organization, covering technical query resolution, estimation, development, integration, testing, and final acceptance. The process applies to all team members, including Project Managers, OTS Leads, Engineers, Quality Engineers, and Subject Matter Experts (SMEs).

3 Abbreviations

a. P&ID – Piping and Instrumentation Diagram

b. OTS - Operator Training Simulator

c. FDS - Functional Design Specification

d. PAM - Process Area Model

e. MAT - Model Acceptance Testing

f. RAIL - Requirements Allocation and Interface List

g. DCS - Distributed Control System

h. SCS – Software Control System

i. ESD – Emergency Shutdown System

j. FAT - Factory Acceptance Testing

k. HMB – Heat and Mass balance

l. KOM – Kick off Meeting

m. MATP- Model Acceptance Test Procedure

n. FATP – Factory Acceptance Test Procedure

o. SATP – Site Acceptance Test Procedure

4 Terms and Definitions

a. Process Area Model (PAM): A detailed representation of a plant section used for simulation.

b. Markup P&ID: A P&ID annotated to indicate changes or specific training model requirements.

c. Punch Fixing: The activity of resolving observations identified during acceptance testing.

d. QC Format: A predefined checklist or document used for quality control of project deliverables.

e. Technical PM and PM are used synonymously in this document.

5 Responsibilities

a. Technical Project Manager:

- Review the agreed proposal and check the outlined scope of the OTS project. Perform a revalidation if required.
- Review the list of deliverables (services, documentation, project milestones, list of software and hardware deliverables.
- Follow up customer/OEM/management for necessary data/documents, software, hardware, VEP's etc
- Review the proposed areas designated as "areas of simplified modeling" and "areas of detailed modeling" with OTS lead.
- Check the fidelity levels proposed to the customer for various modeling sections present it to customer during initial stage and get it approved
- Check the documentation proposed as deliverables to the Customer.
- Prepare and present project execution methodology
- Prepare and present Kick-Off Meeting Agenda to the customer
- Agree upon with the Customer post KOM (Refer FDS for Template)
    - The highlighted scope of the OTS on marked up P&ID's.
    - Field Operated Devices

- Malfunctions
- Instructor Variables
- Boundaries - Process and Utility

- Prepare, present and freeze on the Overall Project Schedule. Ensure to check milestone of Schedule match with Project Proposal Specs
- Brief the customer on the Acceptance Criteria and Model Fidelity/Performance Guarantee
- Demonstrate the Instructor Graphics / Field Graphics Standards to the customer including custom shapes/templates and faceplates and get agreement for OTS.
- Discuss the training requirement with the customer which includes:
  - Type and content of Training
  - Location of Training
  - Schedule of Training
- Review Detailed Functional Specification (DFS) and getting approval end user / OEM
- Share data register to customer/OEM and communicates impact of missing data and assumptions
- Discuss and get agreement on UAT documents for various phases
- Discuss with System (DCS) Engineer on system architecture and requirement specifications
- Approve/ Communicate estimation prepared by system/DCS engineer with customer and get agreement on deliverables
- Project closure meeting with signed copies of completion certificate from end use
- Handover of as built documentation, system maintenance manual handover

Commercial:

- Provide weekly/monthly progress report as per communication protocol for the project
- Follow up for payments as per agreed payment schedule
- Review engineering plan and adjust/communicate/align with delivery dates

b. OTS Lead:

   a. Review highlighted scope of the OTS on marked up P&ID's prepared by modelling engineers.

b. Support PM preparing technical justifications for simplified modeling/specific modeling approach for agreement with customer.

c. Report and support PM in estimation of any additional scope/activities during project which are not in scope as per initial proposal.

d. Review and guide modelling engineers to prepare: (Refer FDS/Detailed SIPOC) for Template)

   - The highlighted scope of the OTS on marked up P&ID's.
   - Field Operated Devices,
   - Malfunctions
   - Instructor Variables
   - Boundaries - Process and Utility

e. Prepare the FDS referring the following documents:

   - Customer specification (RFQ)
   - Correspondence emails and documents
   - Kick-off meeting minutes
   - Marked-up P&IDs
   - Template of FDS
   - Specific FO or customer documentation standards

- Prepare/Update modelling standard document for project if required
- Define Sub Model scope for the in-scope areas
- Prepare Engineering Plan for distribution of Sub Model to Project Team
- Perform gap analysis for all available project team and reshuffle the sub models if required
- Provide inputs to modelling engineer documented in "Sub model QC Checklist.xls" Tab "Inputs-To Modelling Engineer"
- Review/Quick check data register prepared by modelling engineers
- Shares project specific standard File and folder structure
- Based on the Chemical Process involved in the simulation, select the Physical Properties package and the components and the same will be used as a starting point for all Engineers working on the Project. Verify the base case if it is provided by customer.

- Critical units/unit operations need to be built in the OEM software to match the H&MB.
- Prepare overall unit (integrated model) HMB deviation and discuss with customer
- Discuss and get agreement with the revised conditions
- Preparing UAT documents for various phases
- Lead the UAT
- Provide expertise on UAT observation solution to the best possible extent
- Coordinate with systems and graphics team for understanding of the scope
- Take handover of the hardware, software, graphics from the system team
- Coordinate for graphic building activities
- Prepare project closure documents and system maintenance manual
- Conduct share and learn sessions
- Share internal and external project progress as per project agreement during KOM

c. Modelling Engineers:

- Configure the sub model using project modelling specific standard, base case, fragments, marked-up P&IDs and customer provided datasheets
- Proactively report and list any missing data/observations/technical queries using the Query Log
- Verify the model results for steady state as per H&MB frozen during the design stage using stream report deviation sheet
- Support project lead justifications for the deviation
- Conduct dynamic testing, start-up and shutdown tests on the model
- Test Instructor graphics and ensure layout, FOD, Process upsets communication with simulator.
- Add MF's, IV's & check generic MF's
- Complete Sub model Checklists covering the following forms:
    - Checklist for Sub-Model Build
    - Sub model Integration Point List,
    - Sub model Outstanding Issues and Assumptions,
    - Data Void

- Sub model Test log and then Submit model for QA.

- Correct any QA observations & model deficiency, if any
- Integrate sub models using integration checklist and achieve a stable integrated model
- Prepare dynamic initialize case for the integrated model
- Support project lead to conduct start-up and shutdown tests on the model using customer provided documents
- Ensure submission of assigned PAM with agreed deliverables with Quality and within assigned schedule
- Participate in client model review for acceptance test
- Record, analyze, document model deficiency in UAT observation log
- Implement observation log and provide closure remarks
- Integrate simulator with DCS by using standard protocol for various simulators
- Test the communication between simulator, graphics and DCS/ESD system on the target hardware
- Updating as built documentation, system manuals
- Conducting share and learn session

d. Quality Engineers:

- Conduct additional quality verification based on defined QC formats.
- Verify deliverables such as FDS, MATP, FATP, SATP with project Lead
- Responsible for Quality Audit according to Internal Quality Checklist available in IMS.
- Support simulation engineers to prepare reasonable responses for deviation

e. Subject Matter Experts (SMEs) (Optional)

- Help improve business performance by identifying problematic areas and providing relevant technical solutions.
- Evaluate the business processes and operations by implementing their subject expertise.
- Offer consulting services in managing business operations, contribute to specialized knowledge, and provide mentorship.

- Help identify whether the technical solutions the mentors provide meet the company/customer requirements.
- Verify technical information, including the training manuals, user guides, and system requirements.
- Publicize various important information and document processes to all stakeholders and board members.
- Ensure that the information has been represented accurately and per the subject's expertise before delivering the solutions to end-users.

f. Estimation Engineer

- Prepare estimates using a standard estimation sheet within submission timeline
- Collaborating with lead, PM and sales to prepare solutions in line with RFQ/SOW.
- Analyze project specifications, and other documents to determine labor, material, and time requirements. Gathering data on material prices, labor costs, and project timelines.
- Supporting the preparation of bid proposals and presentations
- Considering risk assessments and health and safety requirements
- Participating in project reviews and providing insights on cost-saving opportunities
- Preparing cost reports and maintaining records of estimates and project documentation

g. System Lead Engineer (DCS)

- Study hardware and network to set up DCS hardware and build for OTS
- Understand scope and prepare estimation which covers graphic development, testing & QA and submit to PM
- Support project lead to get approval for sample graphics, shapes & faceplates from OEM and/or from end client.
- Prepare graphics standard for project including resolution fitment for the project and share it to PM for approval
- Perform scoping analysis according to markup P&ID's and DCS CI's for OTS
- Communicate with OEM if certain DCS blocks are not getting translated and get approval from the customer for any workaround.
- Build communication between DCS and the simulator using Toolkit provided by Simulator and guides for DCS translation.

- Prepare and update translation inputs, scoping, translation outputs, USO file with no configuration error, architecture and maintenance guide and hand over to the OTS Lead

h. System (DCS) Application Engineer/Graphic Engineer

- Develop instructor graphics as per project specific guidelines
- Co-ordinate with project team for the project specific changes based on markup P&ID's and instructor graphics/Field Graphics

6 RACI

Process	Responsible	Accountable	Consulted	Informed
Review proposal, estimation, PO/WA, RFQ/Requirement Document	PM	OTS Lead	Modeling Engineer	DCS/ System Lead, Customer
Assess modeling scope (simplified vs. detailed)	PM	OTS Lead	Modeling Engineer	DCS/ System Lead, Customer
Prepare execution methodology presentation & documentation	PM	OTS Lead	Modeling Engineer	DCS/ System Lead, Customer
Prepare R&O/RAIL document	PM	Modeling Engineer	OTS Lead	Customer
Mark P&IDs	PM	Modeling Engineer	OTS Lead	Customer
Identify special modeling needs	PM	Modeling Engineer	OTS Lead	Customer
Discuss scope, validation methods, and architecture	PM	OTS Lead	Modeling Engineer	DCS/ System Lead, Customer

Finalize data requirements and cut-off dates	PM	OTS Lead	Modeling Engineer	DCS/ System Lead, Customer
Review and agree on fidelity and performance criteria	PM	OTS Lead	Modeling Engineer	DCS/ System Lead, Customer
Confirm training needs and schedule	PM	OTS Lead	Modeling Engineer	DCS/ System Lead, Customer
Maintain a risk register	PM	OTS Lead	Modeling Engineer	DCS/ System Lead, Customer
Prepare component slate using HMB	OTS Lead	Modeling Engineer	–	PM, Customer
Develop model using PFDs	OTS Lead	Modeling Engineer	–	PM, Customer
Create deviation sheet and get customer approval	OTS Lead	Modeling Engineer	–	PM, Customer
Maintain modeling guidelines	OTS Lead	Modeling Engineer	–	DCS/ System Lead, Customer
Document PAM allocation	OTS Lead	OTS Lead, PM	Modeling Engineer	DCS/ System Lead, Customer
Configure sub model with inputs	Modeling Engineer	Modeling Engineer	OTS Lead	DCS/ System Lead, Customer
Deliver simulation cases	Modeling Engineer	Modeling Engineer	OTS Lead	DCS/ System Lead, Customer
Complete QC checklist and audit tool	Modeling Engineer	Modeling Engineer	OTS Lead	DCS/ System Lead, Customer

Update worksheets and logs	Modeling Engineer	Modeling Engineer	OTS Lead	DCS/ System Lead, Customer
Review interconnections and schedule impacts	PM	OTS Lead	Modeling Engineer	DCS/ System Lead, Customer
Guide engineers on integration methods	PM	OTS Lead	Modeling Engineer	DCS/ System Lead, Customer
Submit deviation matrix and incorporate customer feedback	PM	OTS Lead	Modeling Engineer	DCS/ System Lead, Customer
Prepare MATP and initialize dynamic models	OTS Lead	Modeling Engineer	—	PM, Customer
Estimate and plan graphic development	DCS/ System Lead	Modeling Engineer	OTS Lead	PM, Customer
Develop graphics using DCS standards	DCS/ System Lead	Modeling Engineer	OTS Lead	PM, Customer
Conduct internal QA and review	DCS/ System Lead	Modeling Engineer	OTS Lead	PM, Customer
Finalize and hand over graphics	DCS/ System Lead	Modeling Engineer	OTS Lead	PM, Customer
Ensure hardware/software readiness	PM	OTS Lead	Modeling Engineer	DCS/ System Lead, Customer
Migrate models to Integration environment	OTS Lead	Modeling Engineer	—	PM, Customer

Verify model scope, graphics, and conditions	OTS Lead	Modeling Engineer	–	PM, Customer
Conduct internal MAT and resolve observations	OTS Lead	Modeling Engineer	–	PM, Customer
Conduct pre-MAT and MAT with documentation	OTS Lead	Modeling Engineer	–	PM, Customer
Maintain observation logs	OTS Lead	Modeling Engineer	–	PM, Customer
Address punch points and finalize remarks	OTS Lead	Modeling Engineer	–	PM, Customer
Secure customer approval	OTS Lead	Modeling Engineer	–	PM, Customer
Setup system and translate configurations	DCS/ System Lead	Modeling Engineer	OTS Lead	PM, Customer
Integrate IO and prepare reference files	OTS Lead	Modeling Engineer	–	PM, Customer
Conduct internal FAT and submit reports	OTS Lead	Modeling Engineer	–	PM, Customer
Resolve issues and obtain customer sign-off	OTS Lead	Modeling Engineer	–	PM, Customer

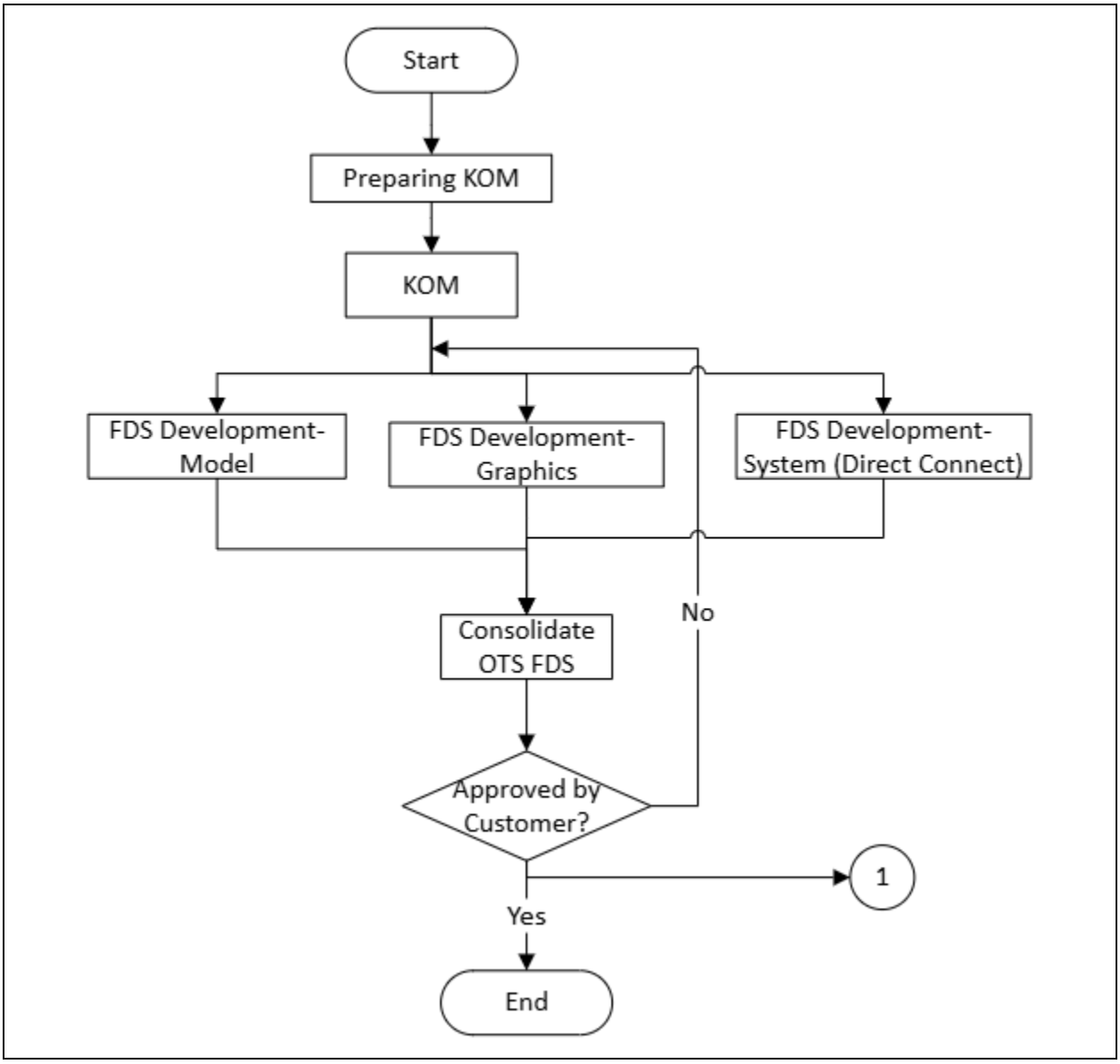
7 SIPOC

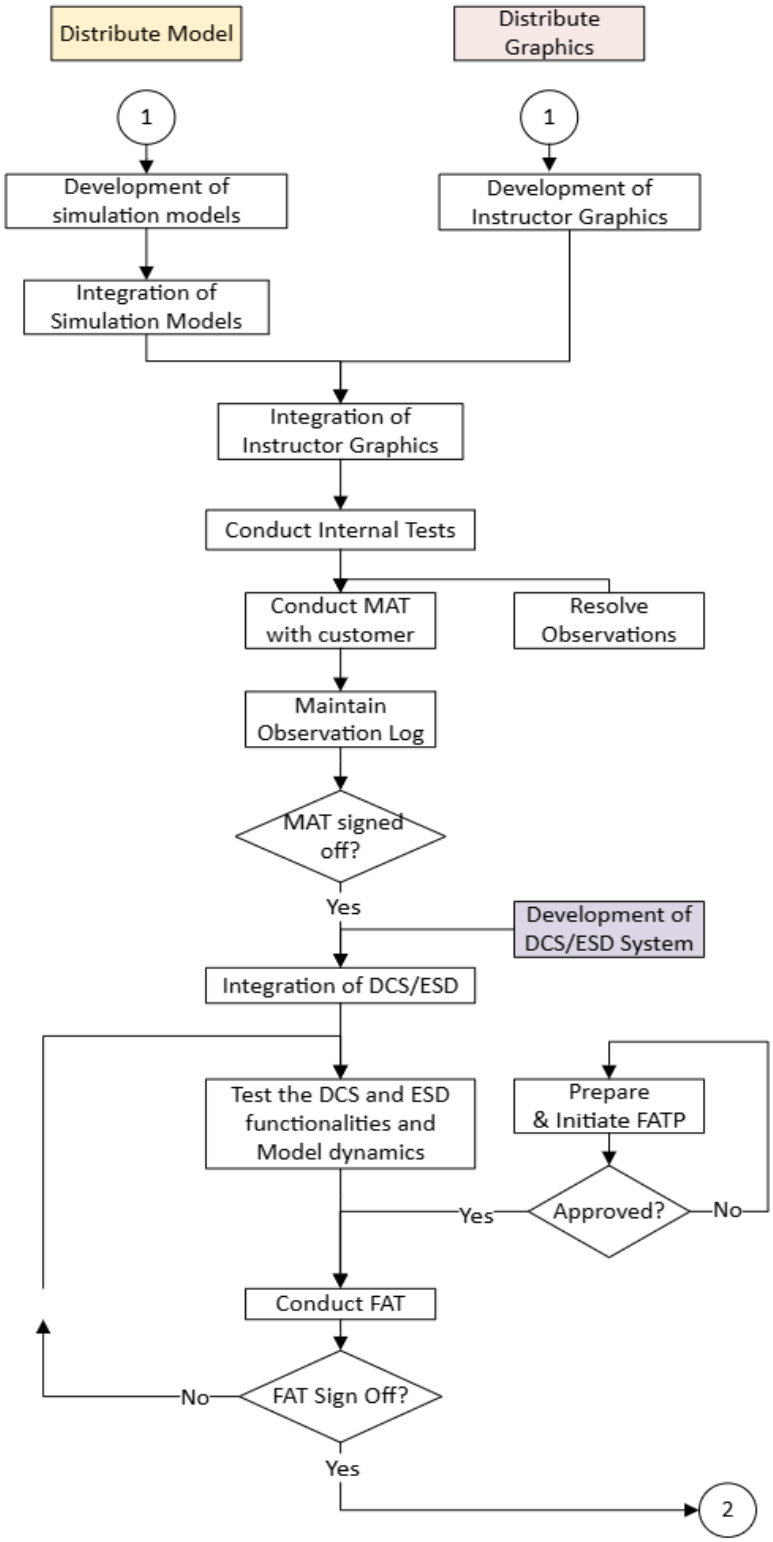
Supplier	Input	Process and Control		Output	Customer
		Process	Control		
Sales Team, Customer	Approved proposal, approved estimation sheet, PO/WA	Pre- Kick off Meeting	Sales to delivery handover checklist, contract review	Scope, Overview of OTS, KOM Agenda, Key Milestones, acceptance criteria, Training plan, system architecture, marked up P&ID's, data register	Customer, BU Head, Quality
OTS Lead	Marked up P&ID's, data register, Kick off deck	Kick off Meeting	Project Kick off checklist Detailed SIPOC.xls	Model validation method, Modeling methodology, FDS, Risks, Resources, agreed deliverables, Signed MOM	PM/LE, Customer, Quality
PM/LE, Customer	Model validation method, Modeling methodology , FDS, Risks, Resources, agreed deliverables, Signed MOM	Design	Data Register, Internal Quality checklist	Data register, Simulation Design Data Management Sheet, Marked up P&ID, PFDs, Graphics, DCS Hardware architecture, Backup files	PM/LE, Customer, Quality

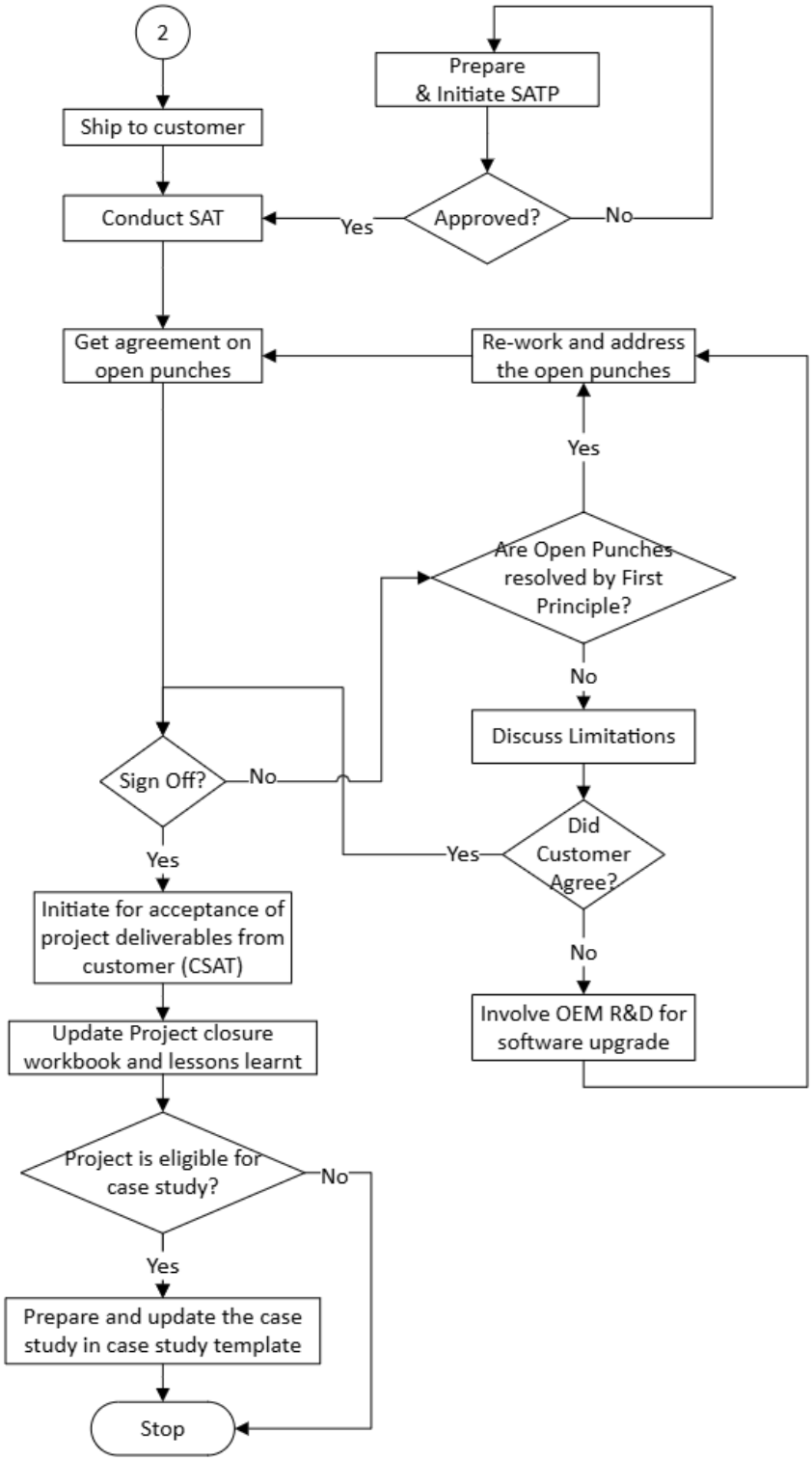
OTS Lead/PM	FDS, Data register, Simulation Design Data Management Sheet, marked up P&ID, PFDs, Graphics, DCS Hardware architecture, Backup files	Development	Data Register, Internal Quality checklist	PAM Model	Application engineer, Customer
Application Engineer	PAM Model	Integration	Internal Quality checklist	Integrated Model	OTS Lead
OTS Lead	Integrated Model, Marked up P&ID, PFDs, Graphics and MATP	MAT	Observation Log	Approved Model, Customer Signed MATP	Customer
OTS Lead	Approved Model, DCS Database, ESD Database & Third-Party Database, FATP	FAT	DCS_ESD Data Requirement Checklist	Integrated Model, Customer signed FATP, Observation log	Customer
PM, Customer	Integrated Model/ Hardware	Shipment	KOM MOM	Delivery of integrated model/hardware	Customer

OTS Lead	Integrated Model/ Hardware, Customer signed FATP, Observation log	SAT	KOM MOM	Integrated model/ hardware, customer signed SATP	Customer
Customer	Customer signed SATP, Customer acceptance certificate	Project Closure	Customer acceptance certificate	CSAT, Lessons learnt, case study	OTS Lead, PM, BU head, Quality

8 Overall Process Flow Chart




Distribute Model
Distribute Graphics
1
1
Development of simulation models
Development of Instructor Graphics
Integration of Simulation Models
Integration of Instructor Graphics
Conduct Internal Tests
Conduct MAT with customer
Resolve Observations
Maintain Observation Log
MAT signed off?
Yes
Development of DCS/ESD System
Integration of DCS/ESD
Test the DCS and ESD functionalities and Model dynamics
Prepare & Initiate FATP
Approved?
Yes
No
Conduct FAT
FAT Sign Off?
No
Yes
2


2
Ship to customer
Conduct SAT
Prepare & Initiate SATP
Approved?
Yes
No
Get agreement on open punches
Re-work and address the open punches
Yes
Are Open Punches resolved by First Principle?
No
Discuss Limitations
Sign Off?
No
Yes
Did Customer Agree?
Yes
No
Initiate for acceptance of project deliverables from customer (CSAT)
Involve OEM R&D for software upgrade
Update Project closure workbook and lessons learnt
Project is eligible for case study?
No
Yes
Prepare and update the case study in case study template
Stop

9 Process Description

This SOP should be read in conjunction with the SOP – Project Management Industrial Solutions (UT/IS/SOL/SOP), which outlines the overall workflow and project-level processes.

9.1 Planning

9.1.1 Project Initiation

Gather critical input documents including:

- Agreed Proposal (Customer/Utthunga/TOC)
- RFQ/Requirement document received from customer
- Estimation Sheet
- Purchase Order (PO) / Work Authorization (WA)

The Technical Project Manager (Technical PM) leads this phase, ensuring scope alignment and preparing key initiation documents such as the KOM Agenda, Organization Chart, and Communication Protocol.

9.1.2 Scope Review & Risk Identification

- Technical PM reviews and validates the project scope.
- OTS Lead identifies simplified, custom, standard modeling areas based on PFDs and P&IDs.
- Prepare Risks & Opportunities (R&O) / RAIL document to proactively manage project risks and opportunities.

9.1.3 Documentation and Deliverable Planning

Define and document key project deliverables, including:

- Modeling Methodology
- System Architecture
- Acceptance Criteria
- Training Plan if applicable
- Preliminary P&ID and Scope Markup

9.1.4 Modeling Preparation

- OTS Lead and Modeling Engineers collaborate to mark up P&IDs and define modeling areas.

- Modeling tasks are supported by engineers, ensuring compliance with OTS-specific modeling standards.
- Functional Design Specification (FDS) and related annexures are prepared and reviewed.

9.1.5 Status Monitoring and Reporting

- Use a Project Initiation Checklist to track the completion status of planning activities.
- Monitor deliverable progress using a Deliverable Responsibility Matrix.
- Visual dashboards (pie charts, bar graphs) are embedded in the project workbook to provide at-a-glance status summaries.

9.1.6 Planning Tools

The planning process is supported by a structured Excel workbook consisting of:

- RACI Chart: Defines task responsibilities.
- Project Initiation Checklist: Tracks task completion.
- Deliverable Responsibility Matrix: Monitors deliverables, owners, and deadlines.

9.2 KOM

9.2.1 KOM Agenda

a. Project Introduction and Objectives alignment

b. Review of Agreed Proposal, Estimation Sheet, and PO/WA

c. Project Scope Review and Deliverable alignment

d. Project Organization Chart Presentation

e. Communication Protocol and Reporting Structure

f. Modeling Methodology and System Architecture Overview

g. P&ID Markup review

h. Simplified, custom and detailed modeling areas agreement

i. Review of Risks & Opportunities (R&O) / RAIL Document

j. Acceptance Criteria and Training Plan Discussion

k. Key Milestones and Project Schedule Overview

l. Q&A Session

9.2.2 Key Roles and Responsibilities

Responsibilities for the project are assigned as follows based on the RACI chart and planning process:

a. Technical Project Manager (Technical PM): Responsible for project scope verification, deliverable management, risk assessment, and overall project coordination.

b. OTS Lead: Responsible for modeling preparation, P&ID markup, FDS preparation, and risk management.

c. Modeling Engineer: Supports modeling tasks, follows standards, assists with documentation, and ensures scope compliance.

d. Customer: Provides input, reviews key documents, and participates in KOM and milestone approvals.

9.2.3 Outcomes

- Project Organization Chart
- OTS Deliverables
- Communication Protocol
- Key Milestones & Activities
- Modeling Methodology
- System Architecture
- Acceptance Criteria
- Training Plan
- Marked up P&ID's and Scope
- Risks & Opportunities / RAIL Document
- Inputs for Functional Design Specification (FDS) and Annexures

9.3 FDS Preparation and Acceptance

9.3.1 Responsibilities

a. OTS Lead

- Define the overall scope and objectives of the OTS project.

- Responsible for Model Scope and Basis including Simulation Models, Basis of Simulation, Thermodynamics, Stream Components, Model Performance, Boundary Conditions, Unit of Measurement, and Naming Conventions.
- Define and validate Initial Conditions, Steady State Condition, and Cold Startup Condition.
- Oversee Training Features including Process Upsets, Instructor Variables, and Field Operated Devices.
- Collaborate on Custom Process Modeling with Modelling Engineers.
- Responsible for Fire and Gas System, Machine Monitoring System, Burner Management System, Compressor/Turbine Anti-Surge Control, Compressor/Turbine Auxiliary System, and Panel Emulation.
- Compile Appendix documentation.

b. System Lead

- Contribute to defining the scope and objectives of the OTS project.
- Responsible for Instructor Graphics/Field Graphics under Training Features.
- Lead Control System Modeling including DCS, ESD, and 3rd Party unsupported PLC Logic.
- Collaborate with Modelling Engineers on Control System Modeling.

c. Modelling Engineers

- Collaborate on Custom Process Modeling with OTS Lead.
- Collaborate with System Lead on Control System Modeling.
- Prepare Appendix documentation including Simulation Drawing List, Equipment List, Unit of Measurement, Process and Utility Boundary List, Custom Process Upset List, Instructor Variables, and Instructor Graphics Shapes/Faceplates.

d. Project Manager

- Contribute to defining the overall scope and objectives of the OTS project.
- Oversee project execution and ensure alignment with goals and timelines.

e. Customer

- Provide input and validation for project scope and deliverables.
- Review and approve key milestones and documentation.

9.3.2 FDS Preparation Process

a. **Step 1. Requirement Analysis**

- Review and analyze business and technical requirements provided by stakeholders. Clarify ambiguities and document assumptions.

b. **Step 2. Functional Breakdown**

- Decompose the system into modules and define the functionality of each component e.g. modelling, DCS, ESD, Graphics etc. Incorporate necessary inputs on graphics and DCS FDS (if it is separate). Use diagrams (e.g., flowcharts, use case diagrams) where applicable. Include them as additional Appendix if applicable

c. **Step 3. Specification Drafting**

- The Specification Drafting phase is the core of the Functional Design Specification (FDS) process. It involves translating analyzed requirements into detailed, structured, and unambiguous functional descriptions that serve as a blueprint for development and testing.
- Key Components of Specification Drafting:
  - Function Identification: Define each function or feature, assign unique identifiers, and group related functions.
  - Purpose and Scope: Describe the objective and boundaries of each function.
  - Inputs and Outputs: List required inputs and expected outputs with format and source/destination.
  - Processing Logic: Detail internal logic using flowcharts, pseudocode, or decision tables.
  - User Interaction: Describe user interface elements and interactions.
  - Dependencies and Interfaces: Identify dependencies and define interfaces and communication protocols.
  - Constraints and Assumptions: Document constraints and design assumptions.

- Error Handling and Exceptions: Specify responses to invalid inputs and system failures.
- Security and Access Control: Define access levels and data protection requirements.
- Validation Criteria: Outline testing and acceptance criteria.

Task	Sub-task
Scope and Definition of OTS	
Model Scope and Basis	
	Simulation Models
	Basis of Simulation
	Thermodynamics
	Stream Components
	Model Performance
	Boundary Conditions
	Unit of Measurement
	Naming Conventions
Initial Conditions	
Steady State Condition	
Cold Startup Condition	
Training Features	
	Instructor Graphics/ Field Graphics
	Process Upsets
	Instructor Variables
	Field Operated Devices
Custom Process Modeling	
Control System Modeling	
	DCS
	ESD
	3rd Party unsupported PLC

Fire and Gas System	
Machine Monitoring System	
Burner Management System	
Compressor/Turbine Anti-	
Compressor/Turbine	
Panel Emulation	
Appendix	
	Simulation Drawing List (In
	Equipment List
	Unit of Measurement
	Process and Utility Boundary List
	Custom Process Upset List
	Instructor Variables
	Instructor Graphics

d. **Step 4. Review and Validation**

- Conduct peer reviews and walkthroughs with relevant teams to validate the accuracy and completeness of the FDS.

e. **Step 5. Version Control and Updates**

- Maintain version history and ensure updates are tracked and communicated effectively.

9.4 Model Building

9.4.1 Responsibilities

a. Technical Project Manager (PM): Oversee project scope, schedule, resource allocation, and customer communication.

b. OTS Lead: Guide modeling activities, review deliverables, and manage quality compliance.

c. Modeling Engineer: Develop, configure, integrate, and validate simulation models in accordance with approved standards.

9.4.2 Procedure

The model building process shall follow the steps outlined below:

a. **Step 1: Review and Preparation**

- Review KOM Minutes of Meeting.
- Obtain approved modeling guidelines, standards, and project-specific annexures.
- Collect inputs: FDS, P&IDs, PFDs, software version, HMB, Data Register, custom libraries (Omegaland), Fragments, smart templates (Unisim) and Equipment Datasheets.
- Create a list of model sections, boundaries, and assumptions.

b. **Step 2: Model Configuration**

- Configure sub-models in alignment with approved standards and scope markups.
- Fix the relevant thermodynamics packages (for unisim Base case+ PRF File) for the model
- Populate process data, equipment characteristics, and control logic.
- Tune the model to match the H&MB
- Validate model sections independently to confirm steady-state operation.

c. **Step 3: Quality Control**

- Complete internal QA checklist for each model section.
- Review exception reports, stream summaries, and pump/compressor curve validations.
- Address deviations in the Critical Point Deviation Matrix and update the Query Log.
- Perform scope check using PID/USD layouts.

d. **Step 4: Integration and Dynamic Initialization**

- Integrate sub-models based on Integration Point Checklist.
- Achieve steady-state operation for interconnected models.
- Submit Integrated Model Deviation Matrix for OTS Lead review.
- Develop Dynamic Initialization (DI)/ Cold conditions model
- Verify dynamic stability and resolve integration issues.

e. **Step 5: Documentation and Handover**

- Prepare final DMS spreadsheets, exception and stream reports, and filled QA checklists.
- Submit completed MATP for customer review.
- Archive the final models (such as (*.utpl), layout printouts, and validated DI models for Unisim) in project repository.
- Conduct model handover and training sessions as per Training Plan.

9.4.3 Checklist

a. **Step 1: Review and Preparation**

- Review KOM Minutes of Meeting
- Obtain modeling guidelines and annexures
- Collect all required inputs (FDS, P&IDs, etc.)
- Define model sections, boundaries, and assumptions

b. **Step 2: Model Configuration**

- Configure sub-models as per standards
- Use Base Case + PRF files for steady-state
- Populate process data and control logic
- Validate each model section independently

c. **Step 3: Quality Control**

- Complete internal QA checklist
- Review exception reports and stream summaries
- Validate pump/compressor curves
- Update Critical Point Deviation Matrix and Query Log
- Perform scope check using PID/USD layouts

d. **Step 4: Integration and Dynamic Initialization**

- Integrate sub-models using checklist
- Achieve steady state for integrated model
- Submit Integrated Model Deviation Matrix
- Develop and verify Dynamic Initialization model

e. **Step 5: Documentation and Handover**

- Prepare final DMS, reports, and QA checklists
- Submit MATP for customer review
- Archive templates, layouts, and DI models
- Conduct handover and training sessions

9.5 PAM Integration

9.5.1 Responsibilities

a. Technical Project Manager (PM): Confirm team size, manage schedule impacts, and customer communication.

b. OTS Lead: Oversee integration planning, model handovers, deviation management, and MATP preparation.

c. Modeling Engineers: Perform integration tasks, steady-state validation, deviation reporting, and DI model creation.

9.5.2 Procedure

a. **Step 1: Resource and Schedule Planning**

- PM to review the project schedule and determine appropriate integration team size.
- Align team availability and resource allocation based on planned integration windows.

b. **Step 2: Sub-Model Handover and Review**

- OTS Lead to formally receive sub-models from Application Engineers.
- Review Integration Point Checklist from Internal Quality Workbook for stream counts and interconnections.
- Assess potential impacts and discuss adjustments with PM.

c. **Step 3: Schedule Impact and Approval**

- PM to notify the customer of any potential schedule or estimate revisions due to integration complexity.
- Obtain customer approval for revised schedule if applicable.

d. **Step 4: Integration Task Allocation and Guidance**

- OTS Lead to distribute integration tasks among Modeling Engineers.
- Provide clear instructions on integration methodology, stream linking, and boundary conditions.

e. **Step 5: Model Integration and Steady-State Validation**

- Modeling Engineers perform sub-model integration ensuring accurate stream interconnections.
- Achieve steady-state operation and resolve model conflicts.
- Submit Integrated Model Critical Point Deviation Matrix to OTS Lead.

f. **Step 6: Deviation Review and Customer Feedback**

- OTS Lead reviews deviations with the customer.
- Record customer inputs or acceptances for deviations.
- Implement customer inputs in the integrated model.

g. **Step 7: Dynamic Initialization and Stability Check**

- Modeling Engineers to configure the Dynamic Initialization (DI) model based on integrated steady-state build.
- Validate DI model for operational stability and dynamic behavior.

h. **Step 8: Documentation and Handover**

- OTS Lead to prepare the Model Acceptance Test Procedure (MATP) covering steady-state and dynamic test cases.
- Archive finalized integrated model files, deviation logs, and QA records.
- Conduct formal handover to customer as per project acceptance protocol.

9.5.3 Checklist

a. **Step 1: Resource and Schedule Planning**

- Reviewed project schedule
- Determined integration team size
- Aligned team availability
- Resource allocation plan

b. **Step 2: Sub-Model Handover and Review**

- Received sub-models from Application Engineers
- Integration Point Checklist from Internal Quality Workbook
- Stream counts and interconnections review

- Impact assessment and adjustments

c. **Step 3: Schedule Impact and Approval**

- Notification to customer of potential schedule revisions
- Customer approval for revised schedule

d. **Step 4: Integration Task Allocation and Guidance**

- Distributed integration tasks among Modeling Engineers
- Instructions on integration methodology
- Stream linking and boundary conditions guidance

e. **Step 5: Model Integration and Steady-State Validation**

- Sub-model integration
- Accurate stream interconnections
- Steady-state operation
- Integrated Model Critical Point Deviation Matrix

f. **Step 6: Deviation Review and Customer Feedback**

- Reviewed deviations with customer
- Recorded customer inputs or acceptances
- Implemented customer inputs in integrated model

g. **Step 7: Dynamic Initialization and Stability Check**

- Configured Dynamic Initialization (DI) model
- Validated DI model for operational stability
- Validated dynamic behavior

h. **Step 8: Documentation and Handover**

- Prepared Model Acceptance Test Procedure (MATP)
- Archived finalized integrated model files
- Archived deviation logs and QA records
- Conducted formal handover to customer

9.6 Graphics Development

9.6.1 Responsibilities

a. System Lead Engineer:

- Responsible for designing and preparing graphics standards and customization requirement based on customer requirements.
- Efforts and requirement analysis

b. Technical Project Manager (PM): Responsible for reviewing budget, communicating with customer for system engineer inputs.

c. OTS Lead: Responsible for reviewing and validating the graphics standard.

d. Graphics Engineering Team: Responsible developing reviewing and updating the graphics.

e. Modelling Engineering Team: Responsible for integrating graphics with simulation models.

9.6.2 Procedure

a. **Step 1. Estimation and Requirements analysis**

- Estimate efforts, software, hardware list.
- Prepare graphics development standard ensure graphics are clear, intuitive, and accurately represent the process or system.
- Prepare standardized symbols and color codes.
- Prepare resolution protocols.

b. **Step 2. Graphics Development**

- Use standardized symbols and color codes.
- Use QC check and validation.
- Maintain consistency in layout and design across all graphics.

c. **Step 3. Integration with Simulation Models**

- Integrate graphics with dynamic simulation models to ensure real-time data representation.
- Validate data flow and synchronization between graphics and simulation models.

d. **Step 4. Review and Validation**

- Conduct thorough reviews of all graphics to ensure accuracy and completeness.
- Validate graphics against training requirements and simulation models.
- Document any discrepancies and implement corrective actions.

e. **Step 5. Version Control**

- Maintain version control of all graphics to track changes and updates.
- Ensure all stakeholders have access to the latest versions.
- Archive previous versions for reference and audit purposes.

9.7 MAT

9.7.1 Responsibilities

a. OTS Lead:

- Conduct pre-MAT testing internally,
- Document results using MATP,
- Maintain observation log,
- Discuss observations with end user, ensure closure remarks, and
- Obtain customer sign-off

b. Modeling Engineer:

- Support pre-MAT testing,
- Address observations through punch-fixing activities and ensure closure remarks.

c. Customer Representative:

- Participate in MAT,
- Review and approve observations
- Perform Tests
- Sign off on MAT

9.7.2 Procedure

a. **Step 1. Pre-MAT Preparation**

- Schedule pre-MAT testing with internal team.

- Prepare MATP (Model Acceptance Test Plan) document.
- Conduct pre-MAT testing internally and document results using MATP.
- Maintain observation log for pre-MAT testing
- Support modelling engineers to resolve

b. **Step 2. MAT Execution**

- Schedule MAT with customer representatives.
- Conduct MAT and document results using MATP.
- Maintain observation log for MAT execution.
- Discuss observations with end user.

c. **Step 3. Observation Handling**

- Address observations through punch-fixing activities.
- Ensure closure remarks for each observation.
- Discuss closure remarks in observation log with end user.
- Obtain agreement from end user on closure remarks.

d. **Step 4. Closure and Sign-Off**

- Ensure all observations are addressed and closed.
- Obtain customer sign-off on MAT.

9.8 Observation Clearance

9.8.1 Responsibilities

a. Observation Owner: Assigned team member responsible for resolution.

b. Reviewer/Auditor: Person who raised the observation.

c. Project Manager: Oversees tracking and closure.

d. Quality Lead: Ensures compliance and documentation.

9.8.2 Procedure

a. **Step 1: Observation Logging**

- Record observation in the Observation Tracker.
- Include: ID, date, description, source (e.g., FAT, SAT), severity, and owner.

b. **Step 2: Categorization**

- Classify as:
  - Critical – Must be resolved before proceeding.
  - Major – Affects performance or safety.
  - Minor – Cosmetic or documentation-related.

c. **Step 3: Assignment**

- Assign to responsible person/team.
- Set target resolution date.

d. **Step 4: Investigation & Resolution**

- Analyze root cause.
- Implement corrective action.
- Document resolution steps.

e. **Step 5: Verification**

- Reviewer verifies resolution.
- May involve re-testing or inspection.

f. **Step 6: Closure**

- Mark as "Closed" in tracker.
- Attach supporting evidence (photos, test reports, updated documents).
- Quality Lead reviews and approves closure.

9.9 DCS Translation (If applicable)

9.9.1 Responsibilities

a. System Lead Engineer: Oversees translation and integration.

b. Simulation Engineer: Implements and tests translated logic.

c. OTS Lead: Validates translation accuracy and performance.

9.9.2 Tools & Resources

a. Honeywell Experion PKS Engineering Stations

b. OEM Software (such as UniSim) Operations Toolkit

c. OPC Communication Setup

d. P&IDs, Control Narratives, DCS CLs

e. OEM Documentation and Support

9.9.3 Procedure

a. **Step 1: Pre-Translation Analysis**

- Review P&IDs and control logic inputs.
- Define simulation boundaries.
- Confirm DCS configuration availability.

b. **Step 2: Extraction of Control Logic**

- Export control modules and tag lists.
- Identify proprietary blocks needing special handling.

c. **Step 3: Translation Mapping**

- Use OEM software (such as UniSim) Operations Toolkit for logic replication.
- Map DCS tags to simulator variables.
- Apply workaround logic for unsupported blocks.

d. **Step 4: Communication Setup**

- Configure OPC DA/UA or custom protocols.
- Validate real-time data exchange.

e. **Step 5: Validation & Testing**

- Perform dry runs and logic verification.
- Validate alarms, interlocks, and HMI graphics.

f. **Step 6: Documentation & Approval**

- Record mappings, workarounds, and test results.
- Submit for customer review and approval.

9.9.4 Troubleshooting Guide for Process Engineers

Issue	Possible Cause	Resolution
DCS blocks not translating	Proprietary or unsupported function blocks	Contact OEM; use equivalent logic or placeholders

Tag mismatch	Incorrect mapping or naming convention	Cross-check tag list; use automated mapping tools
Communication failure	Port issues, firewall, or protocol mismatch	Verify OPC settings; check network configuration
Simulator not responding	Logic errors or missing interlocks	Review control strategy; validate signal flow
Alarm behavior inconsistent	Alarm limits or priorities not replicated	Match alarm settings; test alarm scenarios
HMI graphics issues	Missing or mismatched tags	Update graphic elements; ensure tag linkage
Simulation lag	High data load or inefficient logic	Optimize logic; reduce polling rate

9.9.5 Best Practices for Process Engineers

- Start with a clean, validated DCS configuration.
- Use version control for all logic and mapping changes.
- Maintain a detailed translation log for traceability.
- Automate tag mapping and logic replication where possible.
- Engage OEM support early for proprietary block handling.
- Validate logic, communication, and HMI in separate stages.
- Simulate real-world scenarios including abnormal conditions.
- Ensure compliance with industry standards and regulations.
- Provide platform-specific training to engineering teams.
- Conduct stakeholder reviews for realism and usability.

9.9.6 Common Pitfalls to Avoid

Pitfall	Impact	How to Avoid
Skipping pre-analysis of P&IDs and CLs	Incomplete or inaccurate simulation scope	Always perform thorough scoping before translation
Overlooking proprietary blocks	Simulation logic gaps or failures	Identify and document all proprietary blocks early
Inconsistent tag naming	Communication errors and mismatches	Standardize tag naming conventions across systems
Poor documentation of workarounds	Delays in customer approval and future updates	Maintain clear records of all substitutions and logic changes
Testing only normal operations	Missed edge cases and emergency scenarios	Include startup, shutdown, and fault conditions in testing
Late stakeholder involvement	Misalignment with operational expectations	Engage operations and safety teams during validation
Ignoring simulator performance limits	Lag or instability during training	Optimize logic and data exchange rates for performance
Not backing up configurations	Risk of data loss or rollback issues	Always maintain versioned backups of DCS and simulator files

9.10 System Setup

9.10.1 Hardware Components (If applicable)

The hardware components of a DCS system for OTS typically include:

a. **Control Processors:** Execute control algorithms and manage process data.

b. **I/O Modules:** Interface with simulation such as sensors and actuators.

c. **Communication Network:** Facilitates data exchange between control processors and I/O modules.

d. **Operator Workstations:** Provide user interface for operators to monitor and control the process.

e. **Engineering Workstations:** Used for system configuration, programming, and maintenance.

9.10.2 Software Configuration

The software configuration of a DCS system for OTS includes:

a. Control Software: Implements control logic and algorithms.

b. HMI Software: Provides graphical user interface for operators.

c. Simulation Software: Replicates process dynamics for training purposes.

d. Network Management: Manages communication protocols and network security.

9.10.3 Integration with OTS

Integration of the DCS system with the Operator Training Simulator involves:

a. System Setup: Configuring hardware and software components to replicate the actual control system.

b. Data Translation: Converting real-time process data into simulation data.

c. I/O Mapping: Mapping simulated I/O signals to the DCS system.

d. Debugging and Testing: Ensuring the integrated setup functions correctly.

e. Training Scenarios: Developing realistic training scenarios based on process dynamics.

9.10.4 Typical Workflows

The typical workflows for a DCS system in an OTS environment include:

a. System Setup & Translation: Refer to specific DCS system setup and translation guidelines.

b. Integration: Begin integration activities with necessary inputs and configure I/O connections.

c. Debug & Test: Conduct internal testing, prepare FATP, and perform Factory Acceptance

9.11 DCS Integration & Testing (FAT)

9.11.1 Responsibilities

a. OTS Lead:

- Ensure required hardware and software environments are available and correct.
- Oversee model migration and FAT execution.
- Coordinate issue resolution or escalation.

b. Modeling Engineer:

- Perform model startup and integration activities.
- Conduct step-by-step testing, record observations, and maintain logs.
- Update models per internal and customer feedback.

c. Project Manager (PM):

- Coordinate with customer for external escalations.
- Manage schedule impact if any.
- Manage travel and logistics if applicable.

d. Customer:

- Participate in FAT.
- Review and approve observations.
- Perform Tests.
- Sign Off on FAT.

9.11.2 Procedure

a. **Step 1: Environment Preparation**

- OTS Lead to confirm all necessary hardware systems are operational and correct version of software is installed and tested.

b. **Step 2: Model Migration**

- OTS Lead to load the model into the target setup.

c. **Step 3: Model Startup and Verification**

- Modeling Engineer to perform model startup using project-specific OTS startup procedures.
- Validate the following:
  - Model Scope using P&IDs
  - Instructor Graphics
  - Initial Conditions
  - Execute Plant Start-up
  - Steady State Conditions
  - DCS Graphics Communication (In case of direct connect)
  - Instructor Variables and Custom Malfunctions

d. **Step 4: Internal Factory Acceptance Testing (FAT)**

- Conduct internal FAT covering each of the above verification points.
- Maintain an Observation Log documenting any deviations or anomalies and confirmation of successful verifications.
- Save all intermediate files as per FAT requirements after each test activity.

e. **Step 5: Observation Review and Escalation**

- OTS Lead to review Observation Log, identify issues for internal resolution, and escalate unresolved or customer-dependent issues to the PM.
- PM to manage customer communication for escalations and obtain approvals or guidance for required corrections.

f. **Step 6: Update and Finalization**

- Modeling Engineer to incorporate all inputs and resolutions and update final models accordingly.

g. **Step 7: Sign Off**

- Ensure all observations are addressed and closed.

- Obtain customer sign-off on FAT.

9.11.3 Checklist

a. **Step 1: Environment Preparation**

- Hardware and software environment confirmation document

b. **Step 2: Model Migration**

- Model with all relevant conditions has been transferred.

c. **Step 3: Model Startup and Verification**

- Model startup validation report
- P&ID validation report
- Instructor Graphics validation report
- Initial Conditions validation report
- Plant Start-up execution report
- Steady State Conditions validation report
- DCS Graphics Communication validation report
- Instructor Variables and Custom Malfunctions validation report

d. **Step 4: Internal Model Acceptance Testing (FAT)**

- Internal FAT report
- Observation Log
- Intermediate files

e. **Step 5: Observation Review and Escalation**

- Reviewed Observation Log
- Escalation report

f. **Step 6: Update and Finalization**

- Updated final models

9.12 Shipment

9.12.1 Responsibilities

a. Logistics Team:

- Responsible for coordinating the shipment, including packaging, labeling, and transportation arrangements.

b. Quality Assurance (QA) Team/OTS Lead:

- Conducts pre-shipment inspections and ensures all components meet quality standards.

c. Project Manager:

- Oversees the entire shipment process and ensures compliance with client requirements.

d. Vendor:

- Provides necessary documentation and assists with packaging and labeling as needed.

9.12.2 Pre-Shipment Inspection

- Verify all components of the OTS system are present and in working condition.
- Check for any physical damage or defects.
- Ensure all software licenses are active and valid.
- Confirm all documentation (manuals, certificates, etc.) is complete and up-to-date.

9.12.3 Packaging Requirements

- Use appropriate packaging materials to protect the OTS system during transit.
- Ensure all components are securely packed to prevent movement and damage.
- Use anti-static packaging for electronic components.
- Include desiccants to prevent moisture damage.

9.12.4 Labeling and Documentation

- Label all packages with the following information:
  - Client name and address
  - Project name and reference number
  - Contents of the package
  - Handling instructions (e.g., "Fragile", "This Side Up")
- Include a packing list with each shipment, detailing all components and their quantities.

- Provide all necessary shipping documents (e.g., bill of lading, commercial invoice).

9.12.5 Transportation Arrangements

- Coordinate with the logistics provider to arrange transportation.
- Choose a reliable carrier with experience in handling sensitive equipment.
- Ensure the shipment is insured for its full value.
- Track the shipment and provide regular updates to the client.

9.12.6 Handover Procedure

- Upon arrival at the client site, conduct a handover inspection with the client.
- Verify all components have arrived in good condition.
- Provide the client with all necessary documentation and instructions.
- Obtain client sign-off on the delivery.

9.12.7 Post-Shipment Follow-Up

- Follow up with the client to ensure the OTS system is functioning as expected.
- Address any issues or concerns raised by the client.
- Document any feedback or lessons learned for future shipments.

9.13 SAT

9.13.1 Responsibilities

a. OTS Lead:

- Ensure required hardware and software environments are available and correct.
- Oversee model migration and SAT execution.
- Coordinate issue resolution or escalation.

b. Modeling Engineer:

- Perform model startup and integration activities.
- Conduct step-by-step testing, record observations, and maintain logs.
- Update models per internal and customer feedback.

c. Project Manager (PM):

- Coordinate with customer for external escalations.
- Manage schedule impact if any.
- Manage travel and logistics if applicable.

d. Customer:

- Participate in FAT.
- Review and approve observations.
- Perform Tests.
- Sign Off on FAT.

9.13.2 Procedure

a. **Step 1: Visual & Documentation Inspection**

- Verify physical installation of all hardware components.
- Check wiring, labeling, and mounting against approved drawings.
- Inspect for any physical damage or missing parts.
- HW Bill of Materials (BOM)
- SW Bill of Materials (BOM)
- Compliance documents
- Observation Logs
- Handover documents
- User manuals and installation guides

b. **Step 2. Power-Up Test**

- Ensure system powers up without errors.
- Check boot sequence and software initialization.
- Validate system stability post power-up.

c. **Step 3. Communication Protocol Checks**

- Test integration with control systems (e.g., DCS, PLC).
- Validate data exchange using protocols like Modbus, OPC, Ethernet/IP.
- Confirm real-time data flow and synchronization.

d. **Step 4. Functional Testing**

- Run operational scenarios:
- Normal startup and shutdown if FAT failed
- Emergency shutdown (ESD)
- Fault conditions (e.g., sensor failure, equipment trip)
- Demonstrate Open Observation.
- Validate alarm generation, prioritization, and acknowledgment.
- Demonstrate Instructor variables, malfunctions, scenarios as agreed.

e. **Step 5. Performance Testing**

- Conduct load testing if applicable.
- Monitor system response time and resource usage.
- Ensure consistent performance under simulated operational conditions.

f. **Step 6. Operator Interface Validation**

- Test HMI/SCADA screens for usability and accuracy.
- Validate control actions, trends, and event logging.
- Confirm operator training scenarios are accessible and functional.

g. **Step 7. Logging and Reporting**

- Record all test results and observations.
- Capture screenshots or video logs if needed.
- Prepare SAT report for client review.

h. **Step 8: Observation Review and Sign Off**

- Ensure all observations are addressed and closed.
- Obtain customer sign-off on SAT.

9.13.3 Checklist

a. Visual & Documentation Inspection

- HW Bill of Materials (BOM)
- SW Bill of Materials (BOM)
- Compliance documents
- Observation Logs

- Handover documents
- User manuals and installation guides

b. Power-Up Test

- Power-up test report
- Boot sequence logs
- System stability report

c. Communication Protocol Checks

- Integration test report
- Data exchange validation report
- Real-time data flow confirmation

d. Functional Testing

- Operational scenario test report
- Alarm validation report
- Instructor variables and malfunctions demonstration report

e. Performance Testing

- Load testing report
- System response time logs
- Performance consistency report

f. Operator Interface Validation

- HMI/SCADA usability report
- Control actions validation report
- Operator training scenarios validation report

g. Logging and Reporting

- Test results and observations log
- Screenshots or video logs
- SAT report

h. Observation Review and Sign Off

- Observation log
- Customer sign-off document

9.14 Project Closure

- Technical Project Manager shall follow the Project closure procedure and close the project. The activities include
  - Get project deliverable acceptance from customer
  - Initiate C-SAT survey/collect feedback
  - Capture lessons learnt and best practices followed
  - Update knowledge base with relevant case studies
  - Archive the Project as agreed with the client

10 Information Security requirements

a. Ensure all customer documents and intermediate files are stored securely.

b. Maintain version control for all deliverables.

c. Follow ISO 27001:2022 security guidelines for data handling and storage.

d. Restrict access to sensitive project data

11 References

a. ISO 9001 Clauses 8.1, 8.2, 8.3, 8.5, 8.6, 8.7

b. Risk Management Guidelines UT/PM/PPP/GDL/01

c. SOP- Customer Feedback (CFP)-UT/PS/CFP/SOP

d. SOP- Release, Replication, Delivery and Installation (RRD)- UT/SD/RRD/SOP

e. SOP - Packing & Shipment - UT/PS/PAS/SOP

f. SOP - Project Closure - UT/PM/PCP/SOP

g. SOP – Project Management Industrial Solutions - UT/IS/SOL/SOP

h. SOP - Configuration Management (CMP)-UT/PS/CMP/SOP

i. SOP - Engineering Procurement Construction and Commissioning - UT/EPC/SOP

j. SOP - Customer Feedback (CFP)-UT/PS/CFP/SOP

12 Annexures

12.1 Annexure -1 Honeywell

a. DCS system setup and Translation – TPS Direct connect (Honeywell)

- Inputs for the system setup

- DCS Architecture drawing
- Proposed OTS Architecture drawing
- .lcn
- .cl
- .eb
- DCS Schematics
- Screenshots

- System Lead Engineer shall
  - Study hardware and networking
  - Set up DCS hardware and build for OTS
  - Perform scoping analysis according to markup P&ID's and DCS CI's for OTS
  - Communicate with OEM if certain DCS blocks are not getting translated
  - Communicate workaround done and get it approved from customer
  - Use OEM Software (such as UniSim) Operation Toolkit and guides for DCS translation
  - Build communication between DCS and simulator
- System lead engineer to handover following documents to OTS Lead
  - Translation inputs
  - Scoping
  - Translation outputs (.dcs, .flwio, .ucndef, iomap, .ref)
  - USO file with no configuration error
  - Architecture and maintenance guide (if applicable)

a. **DCS system setup and Translation- EPKS Direct Connect (Honeywell)**

- Input for the System Setup
  - EPKS Version
  - EMDB : Exported all Aseets (*.cnf/*.sl/*.gui)
  - ERDB : Exported all CMs (*.xml/*.sl/*.gui/*.bdc)

- SCADA : Exported QDB files (*.hdw/*.pnt/ or *.xls) for all channels, controllers and points
- OPC Integrator : Exported OPCI file (*.txt)
- DCS Graphics
- HMIWeb graphics (located in the abstract folder) - *.html
- Custom Faceplates (located in the EPKS\client\system folder)
- Custom Detail Displays (located in the EPKS\client\system folder)
- Custom Context Menus (located in the EPKS\client\system folder)
- Custom System Detail displays (display path capture)
- Custom System Detail displays (display path capture)

- System lead engineer shall
  - Study EPKS DCS Requirements
  - Set up EPKS setup (Server/SCE etc) using EPKS DCS Setup Workflow.
  - Set up Customer Database update on OTS setup using standard documents
  - Set up instructor machine using UniSim Instructor Machine Setup Procedure
  - Use UniSim Operation Toolkit and guides for DCS translation
- OEM should support in resolving the issues in timely manner
  - Translation inputs
  - Scoping
  - Translation outputs (.TRAINER.HDW and TRAINER.PNT, . QBDatabase0, iomap, . FSCSMMTABLES)
  - USO file with no configuration error
  - Architecture and maintenance guide (if applicable)

b. **DCS Translation for TPS emulation**

- Input for the Translation Setup
  - .lcn
  - .cl
  - .eb

- DCS Schematics
- Screenshots

- System lead engineer uses UniSim Operation Toolkit and guides for DCS translation
- OEM should support in resolving the issues in timely manner
- System lead engineer to handover following documents to OTS Lead
  - Translation inputs
  - Scoping
  - Translation outputs (.dcs, .flwio, .ucndef, iomap, .ref)
  - USO file with no configuration error
  - Architecture and maintenance guide (if applicable)

c. **DCS Translation for other emulation**

- System lead engineer uses UniSim Operation Toolkit and guides for DCS translation
- OEM should support in resolving the issues in timely manner
  - Translation inputs
  - Scoping
  - Translation outputs (.dcs, .flwio, .ucndef, iomap, .ref)
  - USO file with no configuration error
  - Architecture and maintenance guide (if applicable)

Note- DCS can be emulated in simulation environment also provided control narratives are present.

d. **ESD Translation of Triconex**

- Inputs:
  - Project Files - *.PT2
  - Custom Library Files - *.LT2
  - Triconex Ladder Logic Project Files - *.DPT and *.LPT
  - Microsoft Visual studio 2012 with Intel Visual FORTRAN Composer XX for USO RXXX
  - TriconnexTextExtract.exe

- System lead engineer shall
  - Study Invensys Triconex Requirements
  - Do preprocessing on available PT2
  - Use UniSim Operation Toolkit and guides for ESD translation
  - Do Processing of the Translator Output
  - Compiling a DLL
- Output:
  - dsmdllWithslibs.dll

e. **Translation of third party**

- Integration with third party can be decided by 3 basic rules:
  - Complexity
  - Necessity
  - Maintainability

    For OTS purposes it is highly recommended to discuss and scope out third-party integration

12.2 Annexure – 2- EMERSON

a. System Setup & Translation for Emerson DeltaV and Mimic

- Prerequisites
  - Ensure all necessary hardware and software components are available and installed.
  - Verify network connectivity between DeltaV, Mimic, and OTS systems.
  - Obtain necessary licenses for DeltaV and Mimic software.

b. Setup Steps

- Install DeltaV and Mimic software on the designated systems.
- Configure network settings to enable communication between DeltaV, Mimic, and OTS.
- Set up DeltaV controllers and I/O modules according to the project specifications.
- Configure Mimic simulation environment to replicate the process plant setup.

c. Translation Tools

- DeltaV Configuration Studio: Used for configuring DeltaV controllers and I/O modules.Mimic Simulation Studio: Used for setting up and managing the Mimic simulation environment.OTS Integration Toolkit: Used for integrating DeltaV and Mimic with the OTS system.

d. File Types

- DeltaV Configuration Files (.dcf): Contains configuration data for DeltaV controllers and I/O modules.
- Mimic Simulation Files (.msf): Contains simulation data for the Mimic environment.
- OTS Integration Files (.ots): Contains integration data for connecting DeltaV and Mimic with the OTS system.

e. Integration Notes

- Ensure all configuration files are correctly loaded into DeltaV and Mimic systems.
- Verify communication between DeltaV, Mimic, and OTS systems through network tests.
- Conduct initial integration tests to ensure proper functionality of the combined system.
- Document any issues encountered during integration and resolve them promptly.

f. References

- To be prepared

12.3 Annexure – 3-Yokogawa

a. System Setup & Translation for Yokogawa CENTUM VP and OmegaLand

- Prerequisites

  Ensure the following prerequisites are met before starting the installation and setup:

  - Physical server or virtual machines with adequate CPU, RAM, and storage.
  - Operating System: Windows Server or Linux (as per Yokogawa specifications).
  - Hypervisor: VMware ESXi or equivalent.
  - Software licenses for CENTUM VP and OmegaLand OTS.

- Network Interface Cards (NICs) configured for isolated simulation and control traffic.
- Secure access via SSH or VPN if remote.

b. System Preparation

- Prepare the virtual machines and allocate resources as per Yokogawa guidelines:
  - Create VMs for CENTUM VP HIS (Human Interface Station), FCS (Field Control Station), and OTS.
  - Allocate CPU, RAM, and storage based on system requirements.

c. Software Installation

- Follow these steps to install CENTUM VP and OmegaLand OTS:
  - Install CENTUM VP:
    - Deploy Engineering Station and Operation Station software.
  - Install OmegaLand OTS:
    - Load dynamic simulation models.

d. Communication Setup

Set up communication between CENTUM VP and OmegaLand OTS:

- Integrate with CENTUM VP via PCS.

e. System Validation

Run test scenarios to validate system readiness:

- Test scenarios:
- Startup
- Shutdown
- Fault conditions

Confirm synchronization between simulated process and control actions.

f. Best Practices

- Follow these best practices for successful integration:
  - Use identical graphics and control logic in OTS and real DCS for realism.
  - Maintain version control of simulation models and DCS configurations.

- Schedule regular backups and updates for both systems.
- Enable logging and diagnostics for training analysis.

13 Forms

a. OTS KOM Format

b. Data Register

c. DCS ESD Data Requirement Checklist

d. Detailed SIPOC

e. DMS For HMB

f. Engineering Plan Template

g. Internal Quality Checklist

h. MAT OTS

i. Notes Standardization

j. OTS Modeling standard

k. RAIL Template

l. OTS Specific Startup Procedure

m. Technical Query and Observation Log workbook

n. Utthunga Modelling Guidelines

o. Guidelines for P&ID Markup for OTS

p. Functional Design Specification (FDS)